

TASK 1. CASE STUDY

The following questions are related to the article “*Big data resolving using Apache Spark for load forecasting and demand response in smart grid: a case study of Low Carbon London Project*”.

Make necessary assumptions if needed. Relate your discussion to the curriculum articles.

Additional sources that you have used in your answer shall be listed in your reference list.

a)

Identify data characteristics:

- **Discuss the different datasets used in the case study in this paper with respect to their volume, velocity, variety.**

The data used in the case study was provided by the Low Carbon London (LCL) project, specifically the smart meter data, while additional datasets, including weather data, demographic information, and holiday schedules, were added in the case study.

- **Smart Meter Data:**

Smart Meter data is data collected from advanced electric devices, called smart meter, that measures and record energy usage at regular intervals

In terms of volume, smart meter data should have an extremely high volume as it is generated in a huge amount. In the case study, 5 567 households generated data over a 3-year period, resulting in millions of records. Additionally, an estimation in the US says there are over 135 million utility smart meters, each generating up to 400 megabytes per year. This results in a total of 54 petabytes of generated data annually (Heavy.ai, 2024).

The velocity of smart meter data can be described as high because it is generated and transmitted at frequent, regular intervals, such as every 15 to 30 minutes. This combined with a high number of installed smart meters, would result in a constant flow of data that needs to be ingested, processed, and analyzed in near real-time to enable applications like load forecasting. For example, in the case study, the 30-minute interval data from thousands of households creates a continuous data stream, requiring systems that can handle large-scale, high-speed data ingestion without delays.

In terms of variety, smart meter data is likely to exhibit both structured and semi-structured characteristics. It would primarily consist of structured numerical data, such as kilowatt-hour (kWh) readings and timestamps, which are critical for time-series analysis. However, it is reasonable that it can also include semi-structured metadata, such as information about the smart meter itself such as ID, location or event logs.

- **Weather Data:**

Weather data is data collected from meteorological sources, such as weather stations, satellites, or APIs, that record and provide various information about atmospheric conditions like temperature, humidity, wind speed, pressure, sea level and ground level (OpenWeather, 2024).

In the context of volume, weather data would have a high volume because it is generated continuously across multiple locations. Globally, weather data systems generate immense

datasets when considering the combination of real-time measurements, historical data, and forecasts for different regions. For example, just The European Centre for Medium-Range Weather Forecasts (ECMWF) alone generates 287 terabytes of data every single day (Xweather, 2023).

Weather data demonstrates high velocity as it must be collected, processed, and delivered in near real-time to maintain relevance. Massive computational resources, including supercomputers, are employed to run models quickly, ensuring forecasts are timely for decision-making. Any delay in processing risks rendering the forecast obsolete. (Xweather, 2023)

Weather data can be measured in a variety of ways, by a variety of measurement devices, in a variety of formats. The formats can be structured but are most probably semi-structured as additional attributes are delivered around specific requirements and events. As an example, farmers need upcoming temperature and humidity information, while marine vessels need to track ocean currents and etc. (Explorium, 2023) Additionally, most weather APIs provide weather information as JSON, XML, or HTML format (OpenWeather, 2024). In the case study, the DarkSky API is used to retrieve weather data. Though less commonly, some unstructured data may also exist in the form of descriptive weather forecasts, textual reports, or blob data (pictures of the weather), adding to the complexity.

- Demographic Data:

Demographic data is data collected from sources like census forms, surveys, customer records, or market segmentation tools, that provide insights into characteristics of a population. This includes attributes such as age, income, gender, household size, education, marriage (Hayes, 2024). This data can be used to make socio-economic classifications like the ACORN classification used in the case study.

In the context of volume, demographic data would have a moderate volume compared to weather or smart meter data, as it typically consists of static information that is updated infrequently. However, when covering a large population with detailed segmentation, its size can grow significantly. For example, a dataset representing the demographics of a city like London may contain millions of records detailing household characteristics across various groups.

Demographic data would have a relatively low velocity because it is usually static and updated annually or less frequently. However, certain behavioral aspects of demographics, like purchasing trends or energy usage patterns, could introduce dynamic elements that should be updated regularly.

In terms of variety, demographic data would commonly be structured data, such as numerical or categorical attributes like income, gender, email, age, etc. However, it could also exhibit as semi-structured data, in the form of responses from surveys that may not adhere strictly to predefined schemas but still provide valuable insights. Probably less commonly, demographic data could potentially occur as unstructured data from sources like social media or other feedback forms.

- Holiday Data:

Holiday data is information collected from calendars, national records, holiday APIs, and local databases that provide details about public holidays, festivals, and other culturally significant events. In the case of this case study, bank holiday data, along with the other datasets mentioned above, are combined to enrich analysis and build accurate predictive models.

In terms of volume, holiday data would typically have a low volume compared to datasets like weather or smart meter data because it consists of predefined and static information about holiday dates.

The velocity of holiday data would also be extremely low compared to smart meter data and weather data, since the information changes infrequently. Public holidays are usually announced years in advance, thus making the dataset static for long periods.

In the context of variety, holiday data would usually be incorporated as structured data and semi-structured. It would be structured since it usually includes predefined holiday dates, types, country, day and name, and it may also be semi-structured as holidays APIs usually provides this data in JSON or XML formats (API Ninjas, 2024). Unstructured data could include textual descriptions or blob data representing visualization of the holiday.

- **Why is load forecasting in smart grid in this case study a big data problem?**

Load forecasting in the smart grid in this case study is considered a big data problem due to high volume, high velocity and high variety (Su, Introduction to Big Data).

Smart grids generate massive amounts of data from numerous sensors, meters, and devices. For example, over a year and a 15-min resolution, 1 million smart meters generate data of 2920 TB volume (Ali El-Sayed Ali et al., 2024)

The data is also streamed at a very high speed, necessitating near real-time processing to make timely decisions.

The data is not only voluminous but also diverse. It can include structured data like numerical sensor readings, measurements of energy consumption data, semi-structured data like weather data obtained from web services or APIs, and unstructured data like SMS notifications about tariff, energy use, weather reports etc.

In summary, the combination of high data volume, diversity and velocity makes load forecasting in a smart grid a big data problem. The case study highlights the use of Apache Spark to address these challenges by providing scalable data processing and advanced analytics capabilities.

b)

Frame the problem:

- **What is the business problem the load forecasting application in the case study is addressing?**

The business problem the load forecasting application is addressing can be summarized as:

- Enabling Distribution Network Operators (DNO) to leverage big data analytics to gain insights into consumption behaviors, by predicting electricity demands and thus providing better references for demand response (DR) programs.

- **What can be the potential business benefit of a good load forecasting model?**

The potential business benefit of a good load forecasting model is that it can provide better references to the DR programs, so that the DNO can be more flexible with grid conditions and maintain grid security against overloading and power failure. Thus, providing grid stability, and enhances the reliability of electricity delivery. Other benefits could be that it leads to reduced operational costs for DNOs, and improvements for dynamic pricing strategies (DR-program), such as time-of-use tariffs which can incentivize consumers to shift their energy usage to off-peak times.

- **What kind of task(s) can fit the problem of load forecasting in the case (use the common tasks in Lecture "Predictive Analytics" to frame the business problem)? Is this a supervised or unsupervised method?**

The business problem is about predicting electricity demands and power consumption.

This is an example of a regression task, since we want to predict/forecast the outcome of an event based on previous knowledge (historical smart meter data, weather data and actual consumption). (Su, Predictive Analytics)

This is a supervised method because it involves a set of labeled records (training data) from which the machine learns to predict outcomes for new data. (Su, Predictive Analytics).

c)

Descriptive and predictive analytics:

- **Figure 9 to figure 13 are examples of descriptive analytics the authors proposed. Can you think of one more example of such descriptive figures that can be useful for this case? Describe it.**

The figures 9 to 13 are all examples of descriptive figures that compare average energy consumption per hour (kWh) against various factors such as 24-hour timeframes, months combined with average apparent temperature, apparent temperature alone, months combined with ACORN categories, and ACORN categories differentiated by weekend or working day.

One more example that could be useful for this case is a histogram over the frequency distribution for average energy consumption (kWh) for one household in a week or month. The description of this chart is as follows:

- The x-axis represents the bins or ranges, such as for example from 0.15–0.16 kWh, and so on.
- The y-axis indicates the frequency, specifically the number of households within each bin.

A histogram over frequency distribution would allow DNOs to understand how energy consumption is distributed among households or users. This chart could help in detecting anomalies or outliers, such as households with unusually high or low energy consumption levels. Additionally, it could provide insights into seasonal or temporal patterns in energy usage, such as higher consumption during winter months for heating or summer months for cooling, helping DNOs anticipate and plan for peak load periods effectively.

Assumptions:

- The way the task was framed indicated to me that it may be a specific descriptive figure that is the specific answer. Since the figures from 9 to 13 in the case study include a variety of bar charts, line charts, time series charts, and a scatter chart, I assumed the intent was to propose a different type of chart rather than using the same chart types with different axis descriptions. I considered options like an area chart or radar chart to display energy consumption grouped by the ACORN category, but these did not provide additional insights beyond what was already presented. Instead, I decided on a histogram to show the distribution of average energy consumption per household, as it offers a fresh and potentially more insightful visual interpretation. I am not sure if this was the correct figure if there is a specific answer.
- **This study used Multivariate Polynomial Regression (an extension of Linear Regression) and Decision Tree Regression models for load forecasting. What are the advantages of these models? Why are they (these advantages) important for the smart grid domain?**

Multivariate Polynomial Regression (MPR) and Decision Tree Regression (DTR) are both used in the case study and provide advantages when working with the smart grid domain.

MPR is, as the name suggests, an extension of the classic linear regression (LR) model, with the difference being that it has higher-order terms in the model than normal LR models. The advantages of this model can be further summarized into flexibility, accuracy interpretability and extrapolation (FasterCapital, 2024).

- Flexibility: By adding higher-order terms to the model, it can capture more complex and non-linear relationships between the target and dependent variables, especially when dealing with complex data sets. The ability of MPR to model non-linear relationships is critical in the smart grid domain, where energy consumption patterns are often influenced by multiple factors like time, weather, and socio-economic variables (Ali El-Sayed Ali et al., 2024). Capturing these complexities allows for more realistic and precise load forecasting.
- Accuracy: MPR can also improve the model's accuracy by reducing the bias and variance, compared to normal LR models, where the bias tends to get very high. With a non-linear relationship in the data, the variance also tends to get high with LR (FasterCapital, 2024). However, by using a model with higher-order terms, one can reduce the bias and variance, and thus making the model more accurate which is essential in load forecasting and smart grid domain.
- Interpretability: Even though MPR can be more complex than LR models, they can still be interpreted and visualized, in contrast to more complex models like neural networks (FasterCapital, 2024). This would also allow a data analyst to understand the relationships between variables (e.g., weather and energy consumption). This insight could further support informed decision-making, such as demand-response strategies and tariff adjustments.
- Extrapolation: MPR can also to some extent be used for extrapolation (the process of estimating values outside the range of data). Even though such actions can be risky, MPR models would almost always provide a better estimate than linear models (FasterCapital, 2024). In contrast to the smart grid domain this advantage could be useful for anticipating load under new unseen conditions.

DTR is a commonly utilized algorithm in machine learning for predictive modeling tasks. It works by recursively portioning the dataset into subsets based on the values of input features, creating a hierarchical tree structure. DTR have several advantages, among them being interpretability, non-linear relationships, no feature scaling required, handling of both numerical and category data, and robust outliers. (GeeksForGeeks, 2024).

- Interpretability: One of the most important advantages of DTR is their interpretability. Rules of decision learned by the algorithm are easy to understand and visualize, making it simple to explain (GeeksForGeeks, 2024). This transparency is valuable in domains where interpretability is crucial for decision making, such as in the smart grid system.
- Non-linear relationships: DTR can in likelihood with MPR, capture non-linear relationships between features and target variables. Thus, making it crucial in the smart grid domain as its consumption patterns are often influenced by multiple factors like time, weather, and socio-economic variables.
- No feature scaling required: DTR are not sensitive to the scale of features, meaning that there is no need for feature scaling such as normalization or standardization (GeeksForGeeks, 2024). By that reason, there is no need for further preprocessing, making it faster to implement for large and varied datasets of smart grid domain.
- Handling of both numerical and categorical data: DTR can also handle both numerical and categorical features without the need for preprocessing techniques, making it convenient for dataset with mixed data types, which is also a benefit when it comes to the smart grid, as data comes in a lot of variety with different types.
- Robust to outlier: While MPR are relatively sensitive to outliers, DTR are robust to them. Since DTR partition the feature space into regions based on the values of features, outliers tend to have little to no impact on the overall performance (GeeksForGeeks, 2024). This robustness could ensure reliable load forecasting as outliers could be very common in smart grid due to possible anomalies in energy consumption or measurements.

- **The authors used feature selection before training the model. Why did they do that?**

In the case study, the authors used feature selection before training the model to address specific challenges and optimize their machine learning (ML) models. The primary reasons were to mitigate the heavy matrix computations required by the MPR model, as it demands high computational effort, and to decrease the depth and complexity (pruning) of the DTR model, thereby reducing the risk of overfitting. Reducing the number of input features made the models faster, more robust, more interpretable, and more scalable for big data applications (Ali El-Sayed Ali et al., 2024).

- **Looking at Table 6 (Evaluating metrics for different tested models), which model is the best for “only one block” and which one is the best for “all houses”**

From table 6, the evaluating metrics for the different tested models are listed. For only one block (Block #0), the best model is the Plain regression model as it has the highest R² value, and lowest mean average error (MAE), mean squared error (MSE) and root mean squared error (RMSE). This indicates the best predictive performance compared to the other models for a single block.

For all houses in the LCL project, it is the Ridge regression model that has the highest R² value, lowest MAE, MSE and RMSE. However, it is worth mentioning that the Elastic net model also performs (very) similar to the Ridge regression model. In fact, the R², MAE and RMSE values for these two models are identical. The only difference between them is a thousandth in MSE, making them in practice identical.

d)

Process:

- **Assume that you are a project manager at TrøndEnergi. Based on the knowledge you have gained from this case, can you apply the CRISP model and write a brief data analytics proposal on Load Forecasting for TrøndEnergi (see example on lecture “process, team and privacy”- slide “in class ex. – Review a data analytics proposal”)**

Business understanding

Efficient load forecasting is critical for TrønderEnergi AS to manage the increasing complexity of the smart grid system effectively. Accurate forecasting enables better energy resource allocation, integration of renewable energy sources, and the implementation of demand-response programs. The business problem we aim to solve is ensuring grid stability and operational efficiency while minimizing costs.

We propose developing a machine learning model to predict electricity demand based on historical data, weather conditions, holidays data, and socio-economic variables. This model will address the business needs by:

- Reducing energy wastage.
- Optimizing renewable energy utilization.
- Supporting dynamic tariff structures (e.g., DToU tariffs).
- Preventing grid overload by ensuring accurate load management.

Assume that the cost of over-distribution of electricity (e.g., delivering excess power within the distribution network) is 750 NOK per MWh, and the cost of under-distribution of electricity (e.g., insufficient power delivery to end users) is 1 200 NOK per MWh. On average, TrønderEnergi manages 10 000 MWh of electricity demand daily. Historically, without desired forecasting, the grid experiences a 5% mismatch between actual demand and supply due to errors in load prediction.

Let's also assume that the current situation is that the grid over-distributes 3% and under-distributes 2% of the 10,000 MWh. The daily cost of this will result in:

Over-distribution costs: $0.03 * 10\,000\text{ MWh} * 750\text{ NOK/MWh} = 225\,000\text{ NOK/day}$

Under-distribution costs: $0.02 * 10\,000\text{ MWh} * 1\,200\text{ NOK/MWh} = 240\,000\text{ NOK/day}$

Total daily cost: $225\,000\text{ NOK/day} + 240\,000\text{ NOK/day} = 465\,000\text{ NOK/day}$

With an accurate load forecasting model, it is assumed to reduce the mismatch by at least 60%, leaving the mismatch rate to 2% ($0.05 - 0.60 * 0.05$). This will result in new daily costs with the new over-distributed and under-distributed percentage at 1.2% and 0.8% respectively:

Over-distribution costs: $0.012 * 10\,000\text{ MWh} * 750\text{ NOK/MWh} = 90\,000\text{ NOK/day}$

Under-distribution costs: $0.008 * 10\,000\text{ MWh} * 1\,200\text{ NOK/MWh} = 96\,000\text{ NOK/day}$

Total daily cost: $90\,000\text{ NOK/day} + 96\,000\text{ NOK/day} = 186\,000\text{ NOK/day}$

The savings would then be $465\,000\text{ NOK/day} - 186\,000\text{ NOK/day} = 279\,000\text{ NOK/day}$.

When considering annual savings, it would be: $279\,000\text{ NOK/day} * 365\text{ days} = 101\,835\,000\text{ NOK}$

Let's assume that the cost of implementing this load forecast model is estimated to be around 50 million, and the annual operating costs are 5 million. This will result in an annual profit of

kr 101 835 000
-kr 50 000 000
-kr 5 000 000
kr 46 835 000

Also, after the 50 million investment, TrønderEnergi AS will still have the LF model set up, and it can be applied in the future, having already taken into the investment cost and only having the operational costs going forward.

Data understanding:

To ensure the success of the load forecasting model, we need to evaluate whether the necessary data is available and of sufficient quality. Key data sources include historical electricity consumption, weather data, holiday/event data, and socio-economic variables.

We will assess data completeness, accuracy, and structure, ensuring compatibility with analytical tools and accessibility from central or regional servers/APIs

Findings from this phase will guide data preparation activities, ensuring reliable inputs for model development and minimizing risks of inaccuracies. If critical data cannot be resolved, alternative strategies or project adjustments will be proposed. Alternatively, it should be considered to stop the project.

Data preparation

Next, we will ensure that all relevant data required for the load forecasting model is prepared and ready for analysis. This includes setting up connections to various data sources such as historical electricity consumption, weather data, holiday/event data, and socio-economic variables.

We will decide on the appropriate data storage format, such as structured data for relational analysis or semi-structured formats for large-scale data handling. The data will be cleaned and transformed as part of the ETL (Extract, Transform, Load) process, addressing missing values, outliers, and inconsistencies. If necessary, features will be selected to reduce dimensionality and improve model efficiency.

The dataset will be split into training, validation, and test sets to ensure robust model development. Data will be stored in multiple layers, such as raw, transformed, and augmented formats, with privacy safeguarded using anonymization techniques like tokenization. Aggregation will also be applied where appropriate to protect individual customer privacy while maintaining analytical utility. These steps ensure a reliable and scalable foundation for the load forecasting model.

Modeling

For modeling the load forecasting model, we will apply various Multivariate Polynomial Regression (MPR) models and a Decision Tree Regression (DTR) model to identify which model is the best one

for predicting electricity demand. These models will undergo hyperparameter tuning and cross-validation to optimize its accuracy and prevent overfitting.

Evaluation

In this phase, we will finalize the evaluation of the load forecasting model by assessing its performance metrics, such as R^2 , MAE, MSE, and RMSE, to ensure alignment with TrønderEnergi's business objectives. We will document the model's functionality and verify its output with stakeholders to confirm that they are reasonable and address the business problem effectively.

Deployment

Once evaluated, the model will be integrated into TrønderEnergi's production systems. Software engineers will have to connect the model to real-time data pipelines, enabling it to generate predictions at regular intervals, such as minute-wise or hourly. These predictions will feed into automated systems for dynamic tariff adjustments and grid management (the demand respond programs). The model will be periodically retrained with new data to maintain accuracy and adapt to evolving conditions.

References

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